Notes: Martin & Nagel (JFE, 2022)

Market Efficiency in the Age of Big Data

Contents

1 Big picture

- **Question.** In a world with thousands of potentially relevant predictors, what should "market efficiency" mean? If an econometrician finds strong *in-sample* cross-sectional return predictability, does that really imply risk premia, behavioral bias, or mispricing?
- **Setting.** There are N assets and J observable firm characteristics. Cash-flow growth is linear in those characteristics, but the coefficient vector is *unknown*. Investors are homogeneous, Bayesian, and risk-neutral, and they learn the coefficients from data.
- **Why the paper matters.** Traditional market-efficiency tests effectively assume that investors know the cash-flow forecasting model, or can estimate it almost perfectly. That logic can break down when J is not small relative to N.
- **Core mechanism.** In a high-dimensional learning problem, rational investors shrink coefficient estimates toward zero. That shrinkage is optimal out-of-sample, but it leaves *ex post* predictable components in returns because investors priced assets before future data arrived.
- **Main theoretical result.** If J and N are comparable in size, then standard *in-sample* return-predictability tests reject the no-predictability null with very high probability—even though investors use public information optimally in real time.
- **Key contrast.** Out-of-sample tests behave very differently. An econometrician restricted to the same information investors had in real time cannot earn positive expected returns out-of-sample under the paper's benchmark Bayesian-learning model.
- **Interpretation.** The paper argues that in a high-dimensional environment, *in-sample* anomalies are no longer clean evidence of priced risk or irrationality. The problem is not mainly test-size distortion; the problem is that the usual null hypothesis has weak economic content.
- **Big-data intuition.** More predictors make investors' forecasting problem harder. The econometrician, who sees realized data ex post, has a hindsight advantage over the investors who had to price assets in real time.
- **Simulation message.** With $N = 1000$ and J rising toward 1000, in-sample adjusted R^2 rises sharply and rejection rates of standard Wald tests approach one. Even when researchers test one predictor at a time with multiple-testing correction, false "anomaly discovery" becomes very likely.
- **Sparsity message.** Replacing ridge-type shrinkage with Lasso-type sparsity changes little for the main in-sample conclusion. The central wedge between in-sample and out-of-sample predictability remains.
- **Excess-shrinkage message.** If investors shrink *too much* or impose *too much sparsity*, then out-of-sample predictability appears. So out-of-sample return predictability can be informative about whether investors underuse available information.
- **Empirical application.** Using CRSP stocks and a very large set of predictors built from each stock's own return history (simple returns and squared returns over the previous 120 months), the paper finds a large wedge between in-sample and out-of-sample predictability.

- **Empirical punchline.** In-sample methods recover familiar patterns like momentum, long-run reversal, and momentum seasonality. But out-of-sample portfolio returns are much smaller and often close to zero, so the risk-premium or mispricing component is much smaller than in-sample regressions suggest.
- **Economic takeaway.** In the age of big data, researchers should put much more weight on *out-of-sample* evidence. In-sample predictability by itself is no longer enough to motivate a risk-based or behavioral explanation.

2 Data and measurement (model objects and empirical application)

2.1 Baseline theoretical environment

- Time is discrete, $t \in \{1, 2, \dots\}$.
- There are N risky assets.
- Each asset has a vector of J observable characteristics collected in an $N \times J$ matrix X.
- The paper interprets one period as a **long horizon**—roughly something like a decade.
- The priced claim is a **one-period dividend strip**: asset prices at time t are claims to dividends paid at $t + 1$.

Interpretation. The paper deliberately strips away many complications to isolate the learning problem. The key object is not a full dynamic production economy, but a clean asset-pricing environment in which investors must forecast next-period cash flows.

2.2 Cash-flow process

Dividend growth is

$$\Delta y_t = Xg + e_t, \tag{1}$$

where:

- Δy_t is the $N \times 1$ vector of dividend growth,
- g is the $J \times 1$ vector of unknown coefficients linking characteristics to cash-flow growth,
- e_t is the unpredictable shock.

The maintained assumptions are:

$$e_t \sim N(0, \Sigma_e), \qquad \text{rank}(X) = J, \qquad \frac{1}{NJ}\text{tr}(X'X) = 1. \tag{2}$$

Interpretation. The characteristics span the entire public-information set in the model. The normalization on X keeps the scale of characteristics stable when N and J change.

2.3 Prior beliefs and why shrinkage appears

Investors do *not* know g. Before seeing data, they hold prior beliefs

$$g \sim N(0, \Sigma_g). \tag{3}$$

For tractability, the paper specializes to

$$\Sigma_e = I, \qquad \Sigma_g = \frac{\theta}{J} I, \qquad \theta > 0. \tag{4}$$

- The prior mean is zero: investors do not know *which* characteristics matter.
- The prior variance shrinks with J: this prevents the total predictable component of cash-flow growth from exploding as more predictors are added.
- θ controls the overall amount of predictable variation investors think is plausible.

Interpretation. The paper's point is not that investors mechanically distrust all signals. The point is that an economically reasonable prior says the coefficients cannot be arbitrarily large when there are many candidate predictors.

2.4 Posterior mean as ridge regression

After observing data up to time t, the posterior mean is

$$\tilde{g}_t = \left(\frac{J}{\theta t} I + X'X \right)^{-1} X' \overline{\Delta y}_t, \qquad \overline{\Delta y}_t = \frac{1}{t} \sum_{s=1}^{t} \Delta y_s. \tag{5}$$

The paper also writes this as

$$\tilde{g}_t = \Omega_t (X'X)^{-1} X' \overline{\Delta y}_t, \tag{6}$$

where Ω_t is a shrinkage matrix that downweights principal components with weak signal strength. **Interpretation.** This is exactly the Bayesian version of **ridge/Tikhonov regularization**. Shrinkage is stronger when:

- investors have observed fewer periods (t is small),
- the prior is tighter (θ is small),
- the problem is more high-dimensional (J/N is large),
- or the relevant principal component of X has a small eigenvalue.

2.5 Prices and returns

Because investors are risk-neutral, asset prices are expected next-period dividends:

$$p_t = y_t + \mathbb{E}_t[\Delta y_{t+1}] = y_t + X\tilde{g}_t. \tag{7}$$

Realized returns are

$$r_{t+1} = y_{t+1} - p_t. \tag{8}$$

Interpretation. The paper deliberately sets risk premia to zero in the baseline model. This eliminates the usual joint-hypothesis problem and makes the market-efficiency question cleaner: any predictability that appears cannot be blamed on an omitted risk model.

2.6 Empirical application sample and predictors

The empirical illustration uses U.S. stocks from **CRSP**.

- Stocks are excluded if market capitalization is below the **20th NYSE percentile** or if price is below **$1**.
- To avoid microstructure effects, the most recent month is skipped.
- Predictors are each stock's own **simple returns** and **squared returns** from months $t-2$ through $t-120$.
- This creates **238 predictors** in total: 119 lagged simple returns and 119 lagged squared returns.
- Each month, the dependent variable and all predictors are **demeaned cross-sectionally**.
- Each month, all predictors are also **standardized to unit cross-sectional standard deviation**.
- The panel regression gives **equal weight to each month**.

Interpretation. The authors deliberately choose predictors that were, at least in principle, available to investors throughout the sample. That makes the empirical exercise match the paper's big idea: a large information set can create a wedge between in-sample and out-of-sample predictability even without data mining.

2.7 Empirical estimation windows

- For the full-sample ridge illustration, the panel runs from the beginning of **1971** through **June 2019**.
- The ridge penalty is chosen by **leave-one-year-out cross-validation**.
- For the rolling-window comparison of in-sample and out-of-sample predictability, the paper uses **20-year rolling windows**.
- Given the 20-year estimation window and the 10-year predictor history, the first out-of-sample month is **January 1956**.

Interpretation. The empirical application is designed to compare what looks predictable *in hindsight* with what would have looked predictable *in real time*.

2.8 Portfolio-return objects in the paper

The paper works with three closely related return objects:

- **In-sample portfolio return** r_{IS}: based on OLS-fitted returns from the same sample.
- **Out-of-sample portfolio return** r_{OOS}: based on predictive coefficients estimated from earlier data and applied to later returns.
- **Cross-validated portfolio return** r_{CV}: based on shrunk in-sample forecasts where the penalty is chosen by cross-validation.

Interpretation. The whole message of the paper is about the gap between these objects. r_{IS} contains learning-induced hindsight predictability. r_{OOS} isolates what was truly predictable in real time.

3 Models and empirical specifications (all equations + interpretation)

3.1 Return decomposition under Bayesian learning

A central equation in the paper is the realized return decomposition:

$$r_{t+1} = X(I - \Omega_t)g - X\Omega_t(X'X)^{-1}X'\bar{e}_t + e_{t+1}, \tag{9}$$

where

$$\bar{e}_t = \frac{1}{t}\sum_{s=1}^{t} e_s. \tag{10}$$

This splits realized returns into three parts:

- **Underreaction component:** $X(I - \Omega_t)g$.
- **Estimation-error component:** $-X\Omega_t(X'X)^{-1}X'\bar{e}_t$.
- **New innovation:** e_{t+1}.

Interpretation.

- The first term arises because shrinkage makes investors *underweight* true predictive information in characteristics.
- The second term arises because past noise lines up by chance with characteristics and contaminates posterior beliefs.
- The third term is the new truly unpredictable shock.

Unconditionally, $\mathbb{E}[r_{t+1}] = 0$, but conditional on a given realization of g and the learning history, returns contain predictable components.

3.2 Econometrician's in-sample predictive regression

The ex-post econometrician runs

$$h_{t+1} = (X'X)^{-1}X'r_{t+1}. \tag{11}$$

Under the conventional rational-expectations null, the Wald object

$$h'_{t+1}X'Xh_{t+1} \tag{12}$$

behaves like a χ^2_J statistic, and the normalized test statistic is

$$T_{RE} = \frac{h'_{t+1}X'Xh_{t+1} - J}{\sqrt{2J}}. \tag{13}$$

Under rational expectations, the econometrician expects

$$T_{RE} \Rightarrow N(0,1). \tag{14}$$

Interpretation. This is the usual benchmark behind in-sample tests of market efficiency: if investors already know the true forecasting model, characteristics should not predict returns.

3.3 High-dimensional asymptotics

Instead of the usual "large N, fixed J" logic, the paper studies

$$N, J \to \infty \qquad \text{with} \qquad \frac{J}{N} \to \psi > 0. \tag{15}$$

This is the paper's formalization of a high-dimensional or "big data" setting. The key structural assumption is:

$$\lambda_j\left(\frac{1}{N}X'X\right) > \varepsilon > 0 \qquad \text{for all } j, \tag{16}$$

uniformly in the asymptotic sequence. **Interpretation.** The paper does *not* just assume "many variables." It assumes many *nonredundant* variables. If all extra variables were asymptotically spanned by a small core set, the big-data argument would lose force.

3.4 Main in-sample result

The asymptotic logic of Propositions 3 and 4 is:

- Under the true Bayesian-learning model, T_{RE} is *not* centered at zero.
- It has a positive drift of order $\sqrt{J}$.
- Therefore, for any fixed critical value c_α,

$$P(T_{RE} > c_\alpha) \to 1. \tag{17}$$

The paper further shows that the probability of *not* rejecting the no-predictability null declines exponentially fast with N. **Interpretation.** This is the most important proposition in the paper. In a high-dimensional world, strong in-sample return predictability is perfectly compatible with rational Bayesian pricing.

3.5 Why this is not just a statistical-size problem

The paper is explicit that the large rejection rates are *not* mainly because the econometrician used the wrong asymptotic variance formula. Instead:

- equilibrium asset prices genuinely contain in-sample predictable components,
- those components come from learning and shrinkage,
- and they become quantitatively important when J/N is not small.

Interpretation. The critique is deeper than "use robust standard errors." The paper says the economic null itself is misspecified if it assumes investors know the forecasting model.

3.6 In-sample trading strategy

An in-sample strategy that holds assets with weights proportional to fitted returns is

$$w_{IS,t} = \frac{1}{N} X h_{t+1}. \tag{18}$$

Its return is

$$r^{w}_{IS,t+1} = \frac{1}{N} r'_{t+1} X h_{t+1} = \frac{1}{N} h'_{t+1} X' X h_{t+1}. \tag{19}$$

Interpretation. This shows that the in-sample test-statistic logic and the in-sample "anomaly portfolio" logic are basically the same object. If one is large, so is the other.

3.7 Random-matrix benchmark

To make the theory more concrete, the paper studies a benchmark in which the entries of X are IID with mean zero and unit variance. Then:

- the eigenvalues of $\frac{1}{N}X'X$ converge to the **Marchenko–Pastur** distribution,
- the paper can derive closed-form expressions for the limiting mean and variance of the relevant transformed eigenvalues,
- and it can map those moments into explicit rejection probabilities.

Interpretation. This benchmark translates the theory into something quantitative. It shows that the asymptotic logic is not just abstract high-dimensional math; it bites at realistic sample sizes.

3.8 Out-of-sample return-predictability test

The out-of-sample portfolio is

$$r_{OOS,t+1} = w'_{OOS,s+1} r_{t+1}, \qquad w_{OOS,s+1} = \frac{1}{N} X h_{s+1}, \tag{20}$$

with $s \neq t$ so that the estimating sample and evaluation sample do not overlap. The paper's key out-of-sample result (Proposition 6) is:

$$\mathbb{E}[r_{OOS,t+1}] = 0. \tag{21}$$

Interpretation. Once the econometrician is put on the same information footing as investors, return predictability disappears in the benchmark model. This is the sense in which out-of-sample tests retain economic meaning.

3.9 Backward out-of-sample prediction

The paper also studies the case $s > t$, where the econometrician estimates predictive coefficients using *later* data and applies them to *earlier* returns. **Interpretation.**

- This is not a tradable strategy in real time.
- But it is highly relevant for empirical finance, where researchers often revisit older data with a predictor discovered later.
- The model implies that even this backward out-of-sample predictability vanishes under benchmark Bayesian learning.

This gives the paper a neat explanation for why many published anomalies weaken in historical backtests.

3.10 Risk-premium extension

If returns contain an additional expected-return component $X\gamma$, then the model becomes

$$r_{t+1} = X\gamma + X(I - \Omega_t)g - X\Omega_t(X'X)^{-1}X'\bar{e}_t + e_{t+1}. \tag{22}$$

Then the out-of-sample portfolio isolates

$$\frac{1}{N}\gamma' X' X\gamma = \mathbb{E}[r_{OOS,t+1}]. \tag{23}$$

Interpretation. This is a major contribution. Out-of-sample returns reveal the component of expected returns due to risk premia, behavioral distortions, or frictions, because they are not contaminated by learning-induced hindsight predictability.

3.11 Cross-validated penalized regression

The paper also studies the penalized in-sample problem

$$b_t = \arg\min_b (r_t - Xb)'(r_t - Xb) + \xi\, b'X'Xb. \tag{24}$$

The solution is

$$b_t = \frac{1}{1+\xi} h_t. \tag{25}$$

Choosing ξ by cross-validation yields a shrunk portfolio return r_{CV} with

$$\mathbb{E}[r_{CV}] = \mathbb{E}[r_{OOS}]. \tag{26}$$

Interpretation. Cross-validation picks the amount of shrinkage that strips out the hindsight-only component of in-sample predictability. This gives an economic interpretation of machine-learning procedures that tune penalties by cross-validation.

4 Identification and economic interpretation (what makes the claims credible)

4.1 The paper deliberately removes the joint-hypothesis problem

The investors are risk-neutral. So the paper is *not* relying on an omitted risk-pricing model to generate predictable returns. **Interpretation.** If even in this simplified case in-sample predictability can arise mechanically from learning, then the paper's warning about anomaly interpretation is especially strong.

4.2 The paper stacks the deck against finding predictability

The prior is *objectively correct*: the coefficient vector g is actually drawn from the same distribution investors use as their prior. **Interpretation.** This makes the benchmark friendly to market efficiency. The paper is not generating anomalies by assuming investors have the wrong model or irrational beliefs. The main wedge appears even when investors' prior is correctly specified.

4.3 Why in-sample evidence loses economic meaning

The econometrician sees realized data *after* investors priced assets. In a low-dimensional world, that extra information matters little because investors can estimate coefficients precisely. In a high-dimensional world, it matters a lot. **Interpretation.** The econometrician's ex-post regression is not answering the same economic question as the investor's real-time forecasting problem.

4.4 Why out-of-sample evidence keeps economic meaning

Out-of-sample tests restrict the econometrician to the same information set investors effectively had when prices were set. **Interpretation.** That is why the paper repeatedly emphasizes that out-of-sample tests are not merely a robustness check. They test a fundamentally different and more economically meaningful null in a learning environment.

4.5 Why the big-data assumption matters

If the extra predictors are mostly redundant, the usual low-dimensional logic is approximately recovered. The paper's results require the forecasting problem to remain genuinely high-dimensional as N grows. **Interpretation.** The point is not "many columns in a spreadsheet." The point is many pieces of information that investors cannot all estimate with high precision.

4.6 What excess shrinkage or excess sparsity means

The benchmark model chooses shrinkage optimally from a Bayesian point of view. But real investors may:

- face costs of information processing,
- prefer simpler models,
- or use priors that are too tightly concentrated around zero.

Interpretation. In that case, out-of-sample predictability can emerge. So out-of-sample returns become diagnostic of whether investors underreact because they shrink too aggressively.

4.7 Important limitations

- The cash-flow forecasting model is linear.
- Characteristics are fixed over time in the baseline setup.
- Investors are homogeneous.
- The backward out-of-sample result may rely on the paper's particular IID environment.
- The empirical application uses lagged returns rather than truly exogenous characteristics, so it is an illustration rather than a literal structural test.

Interpretation. None of these limitations overturn the main message, but they matter for how broadly one should generalize the exact propositions.

5 Figures and simulations

5.1 Figure 1 and Figure 2: eigenvalues and the rejection-rate geometry

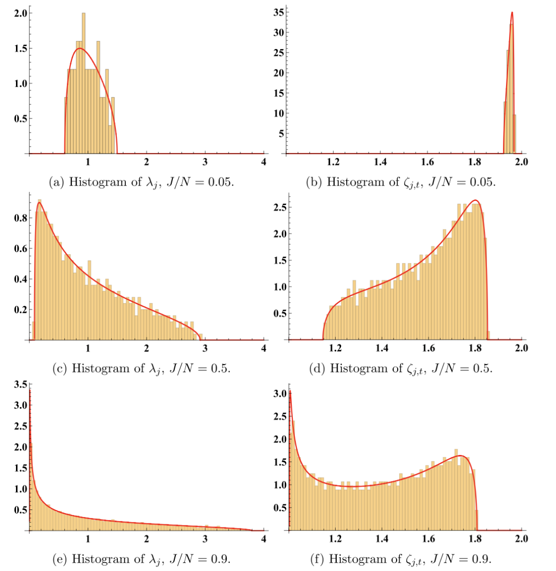

Fig. 1. Histograms of eigenvalue distributions in examples with $\theta = 1$, $t = 1$, $N = 1000$ and $J = 50, 500, 900$. The asymptotic distribution is shown as a solid line in each panel.

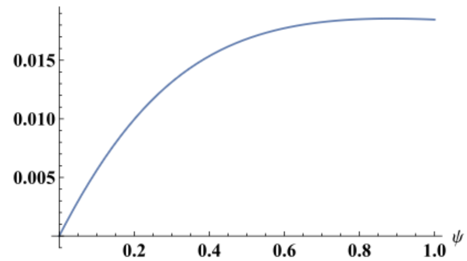

Fig. 2. The rate function given in equation (16).

Read.

- Figure 1 shows eigenvalue histograms for the random-matrix benchmark with $N = 1000$ and different values of J.
- As J/N rises, the distribution places more mass near zero, meaning many predictors are nearly collinear.
- Figure 2 translates the theory into rejection-rate geometry: with $\theta = 1$ and $t = 1$, once $\psi = J/N$ rises above about 0.4, the nonrejection probability becomes extremely small.

Economic message. Even when predictors are noisy and correlated, a genuinely high-dimensional information set makes in-sample rejections almost inevitable.

5.2 Figure 3: in-sample R^2 and rejection rates

Read.

- The simulations fix $N = 1000$ and let J rise toward 1000.
- The adjusted R^2 from in-sample return-predictability regressions rises strongly with J.
- The rise is slower when investors have had more time to learn (larger T), but it remains substantial.
- The rejection probability of the no-predictability null approaches one quickly as J/N rises.
- Even when investors have learned for four periods, rejection rates exceed about 90% once J is more than roughly half of N.

Economic message. Large in-sample anomaly evidence can appear even when investors are fully rational Bayesians. More data for investors helps, but it does not undo the high-dimensional logic.

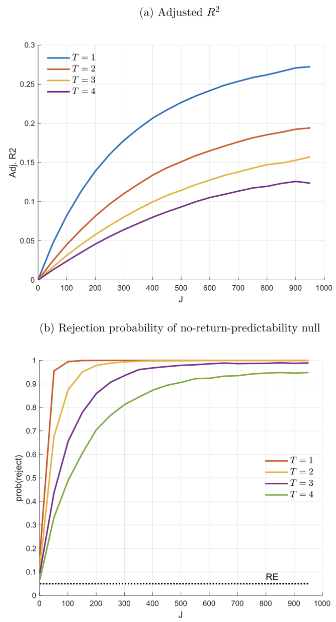

Fig. 3. In-sample return predictability tests. Based on cross-sectional regressions with $N = 1000$ assets and J predictor variables, predicting the last return in a sample of size $T + 1$ and where investors have learned about $\mathbf{g}$ from a sample of size T. The test in panel (b) is a joint χ^2–test using all J predictors. It has an asymptotic 5% rejection probability under the rational expectations null hypothesis (where investors know $\mathbf{g}$). The solid lines show the actual rejection probabilities when this test is applied in a setting where Bayesian investors with an objectively correct informative prior estimate $\mathbf{g}$.

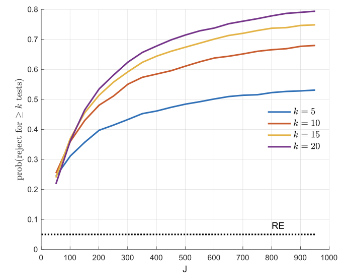

Fig. 4. In-sample return predictability tests with single predictors and correction for multiple testing. Based on J cross-sectional regression with $N = 1000$ assets and one out of J predictor variables in each regression, predicting the last return in a sample of size $T + 1$, and where investors have learned about $\mathbf{g}$ from a sample of size $T = 4$. Critical values are adjusted so that the probability that k of the J hypotheses are rejected would be 5% in the rational expectations case (where investors know $\mathbf{g}$). The solid lines show the actual rejection probabilities when this test is applied in a setting where Bayesian investors with an objectively correct informative prior estimate $\mathbf{g}$.

5.3 Figure 4: single-predictor tests with multiple-testing correction

Read.

- Instead of one kitchen-sink regression, the paper lets many researchers each test one characteristic.
- Critical values are adjusted to control the k-familywise error rate under rational expectations.
- Even with that correction, rejection rates rise sharply as J/N increases.
- For example, when $k = 20$ and $J/N = 0.05$, the actual probability of rejecting at least 20 hypotheses is about 25%, versus the benchmark 5% under rational expectations.
- When J/N is close to one, that rejection probability rises to about 80%.

Economic message. The paper's critique is not limited to big kitchen-sink regressions. A literature full of individually tested predictors can also overdiscover "anomalies" in a high-dimensional learning world.

5.4 Figure 5: Lasso / sparsity version

Read.

- The paper replaces the Normal prior with a Laplace prior and lets investors price using the posterior mode, which corresponds to Lasso-type sparsity.

- The resulting in-sample adjusted R^2 and rejection probabilities look very similar to the ridge/Normal-prior case.

Economic message. The basic wedge between in-sample and out-of-sample predictability is not a fragile feature of ridge regression. It survives when investors use sparse forecasting models.

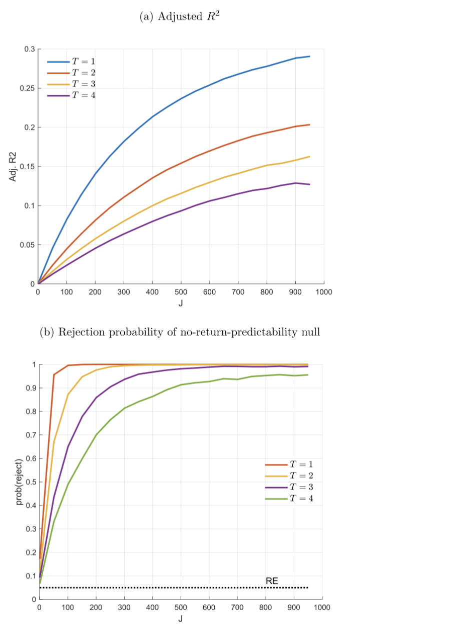

Fig. 5. Lasso: in-sample return predictability tests. Based on cross-sectional regressions with $N = 1000$ assets and J predictor variables, predicting the last return in a sample of size $T + 1$ and where investors have learned about $\mathbf{g}$ from a sample of size T. The test in panel (b) is a joint χ^2–test using all J predictors. It has an asymptotic 5% rejection probability under the rational expectations null hypothesis (where investors know $\mathbf{g}$). The solid lines show the actual rejection probabilities when this test is applied in a setting where investors estimate $\mathbf{g}$ (which is drawn from a Laplace distribution) with Lasso.

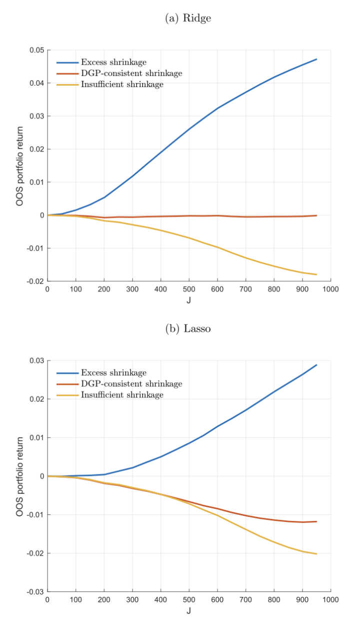

Fig. 6. Out-of-sample portfolio returns when investors apply excess shrinkage or sparsity. Based on cross-sectional regressions with $N = 1000$ assets and J predictor variables, predicting the last return in a sample of size $T + 1$, and where investors have learned about $\mathbf{g}$ from sample of size $T = 4$. Cash-flow data are always generated with $\theta = 1$. In the DGP-consistent prior case, investors' prior is based on $\theta = 1$. In the excess shrinkage (or sparsity) case, investors' prior is based on $\theta = 0.5$. In the insufficient shrinkage (or sparsity) case, investors' prior is based on $\theta = 2$. The DGP in all cases always features $\theta = 1$.

5.5 Figure 6: excess shrinkage and excess sparsity

Read.

- Under *DGP-consistent* ridge shrinkage, average out-of-sample portfolio returns are approximately zero.
- Under *excessive* shrinkage, out-of-sample returns become positive and grow with J.
- Under *insufficient* shrinkage, out-of-sample returns become negative.
- In the Lasso case, the same qualitative pattern holds, though the DGP-consistent case is slightly negative because investors use the posterior mode rather than posterior mean.

Economic message. Out-of-sample predictability can be evidence that investors are not using information in the fully Bayesian-optimal way—for example because they simplify too aggressively.

5.6 Figure 7: ridge coefficients in the empirical application

Read.

- A single ridge regression on 238 lagged return-based predictors recovers several famous anomalies at once.

- Positive coefficients on recent lags capture **momentum**.
- The especially strong positive weight on lags 7–12 matches the **intermediate-horizon momentum** emphasis in the later literature.
- Negative coefficients on longer simple-return lags capture **long-run reversal**.
- Positive coefficients at lags that are multiples of 12 pick up **momentum seasonality**.
- Lagged squared-return coefficients are mostly positive at long horizons, indicating a positive relation between long-run volatility history and future returns.

Economic message. A large, mechanically defined predictor set reproduces many well-known anomalies without the researcher having to cherry-pick a small set of lags.

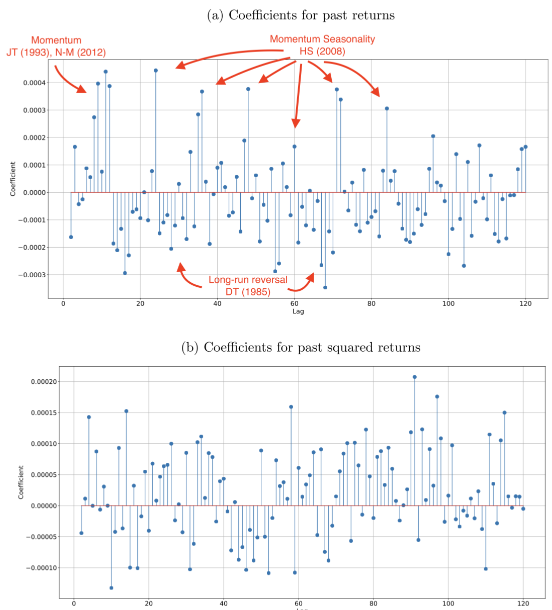

Fig. 7. Ridge regression coefficient estimates.

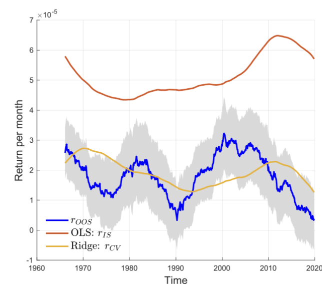

Fig. 8. Rolling estimates of the risk premium/mispricing component of returns. The risk premium component $\boldsymbol{\gamma}'\boldsymbol{X}'\boldsymbol{X}\boldsymbol{\gamma}$ is estimated by the out-of-sample portfolio return $r_{OOS,t+1} = \boldsymbol{r}'_{t+1}\boldsymbol{X}\boldsymbol{h}_t$. We obtain the OLS estimates $\boldsymbol{h}_t$ from regressions of individual stock returns on lagged returns and lagged squared returns in backwards 20-year moving windows up to month t and use them to construct weights applied to month $t+1$ returns. Stocks with market capitalization below the 20th NYSE percentile and lagged price lower than one dollar are excluded. The blue line in the figure shows $r_{OOS,t+1}$ averaged in 10-year moving windows. The shared area indicates two–standard error bands. For comparison, the figure also shows the in-sample return based on cross-validated ridge regression estimates, r_{CV}, and OLS estimates, r_{IS}. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

5.7 Figure 8: in-sample vs out-of-sample portfolio returns

Read.

- The in-sample portfolio return r_{IS} is stably positive over time.
- Its magnitude is about 0.006% per month relative to a monthly cross-sectional return variance around 0.6%, implying an in-sample R^2 of roughly 1%.
- The out-of-sample portfolio return r_{OOS} is much lower, often near zero or below.
- In the 10-year windows ending during 2014–2019, the average out-of-sample return is less than two standard errors above zero.
- The cross-validated in-sample return r_{CV} tracks r_{OOS} much more closely than r_{IS} does.

Economic message. Most of what looks like predictable expected returns in-sample does *not* survive as real-time predictability. Cross-validation largely recovers the out-of-sample component, consistent with the theory.

6 Short summary

- The paper redefines what market-efficiency tests mean when investors face a high-dimensional forecasting problem.
- Bayesian learning plus shrinkage can generate large *in-sample* cross-sectional return predictability, even with fully rational investors.
- Therefore, in-sample anomaly evidence does not by itself identify risk premia or behavioral mispricing.
- Out-of-sample tests remain economically meaningful because they put the econometrician on the same information footing as investors.
- In the benchmark model, out-of-sample predictability is zero.
- If out-of-sample predictability exists, it can reflect either true risk premia / distortions or investors' use of excessive shrinkage / sparsity.
- The empirical application shows a large real-world wedge between in-sample and out-of-sample return prediction when researchers use a big set of lagged-return predictors.

7 Conclusion

7.1 Core conclusion

The paper's core claim is simple but powerful: in a high-dimensional world, the conventional no-predictability null is no longer the right benchmark for market efficiency. Rational investors who learn optimally can still generate returns that look predictable in hindsight.

7.2 Economic intuition

The econometrician has an ex-post advantage because she gets to see future data that investors did not yet have when prices were set. With many potential predictors, that hindsight advantage is first-order. Shrinkage is optimal for investors out-of-sample, but it creates in-sample predictable components in realized returns.

7.3 Takeaway

This paper is best read as a warning against overinterpreting anomaly evidence. In the age of big data, the question is not merely whether returns are predictable *in sample*. The real question is whether they were predictable *in real time*. That is why out-of-sample tests move from being a robustness check to being a central economic test.